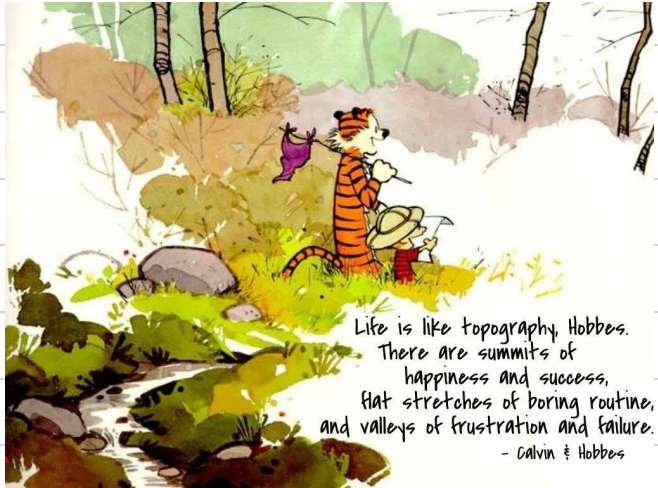


DEAR STUDENTS, ALWAYS
REMEMBER THAT

AT THE END OF
THE DAY, I
SHOULD BE
BETTER THAN
I WAS AT THE
START OF IT

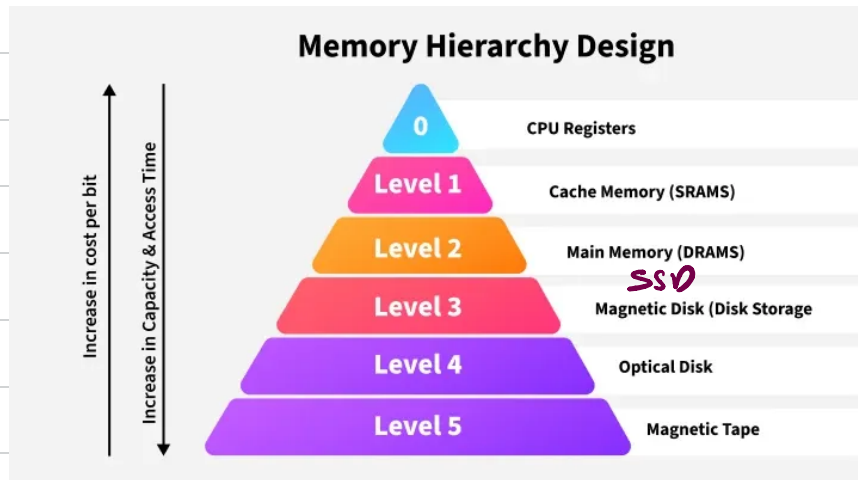
KEEP LEARNING



Issue with Pandas!?

1 GB
→
≫ 1 GB

- ① It tries to load everything in the RAM.
- ② Single Core.
- ③ Does not use distributed computing.



Ryzen 9 7950X3D cache:

Cache	Amount
L1	1 MB / 1024 KB
L2	16 MB
L3	128 MB
Total cache often advertised	145 MB = 1 MB L1 + 16 MB L2 + 128 MB L3

Important detail: the 128 MB L3 is split across two CCDs:

- One 8-core CCD has 96 MB L3: 32 MB normal L3 + 64 MB 3D V-Cache
- The other 8-core CCD has 32 MB L3
- Total: 96 + 32 = 128 MB L3

AMD's official spec lists L1: 1024 KB, L2: 16 MB, L3: 128 MB for the 7950X3D. amd.com

Output:

```
import pandas as pd

# Load a large CSV file in chunks of 10,000 rows
for chunk in pd.read_csv('large_file.csv', chunksize=10000):
    print(chunk.shape)
```

1	(10000, 9)
2	(10000, 9)
3	(10000, 9)
4	(10000, 9)
5	(10000, 9)
6	(10000, 9)
7	(10000, 9)
8	(10000, 9)
9	(10000, 9)
10	(10000, 9)
11	(10000, 9)
12	(10000, 9)
13	(10000, 9)
14	(10000, 9)
15	(10000, 9)
16	(10000, 9)
17	(10000, 9)
18	(10000, 9)
19	(10000, 9)
20	(10000, 9)

[Click to enlarge](#)

<https://api.proxpc.com/media/uploads/2025/01/28/vram-deepseek-table-1.png> .

Model Variant	Parameters	VRAM (FP16)	VRAM (4-bit Quantization)
DeepSeek-LLM 7B	7 billion	16 GB	4 GB
DeepSeek-LLM 67B	67 billion	154 GB	38 GB
DeepSeek V2 16B	16 billion	37 GB	9 GB
DeepSeek V2 236B	236 billion	543 GB	136 GB
DeepSeek V2.5 236B	236 billion	543 GB	136 GB
DeepSeek V3 671B	671 billion	1,543 GB	386 GB

DASK

→ 70% of API is v.
similar to pandas

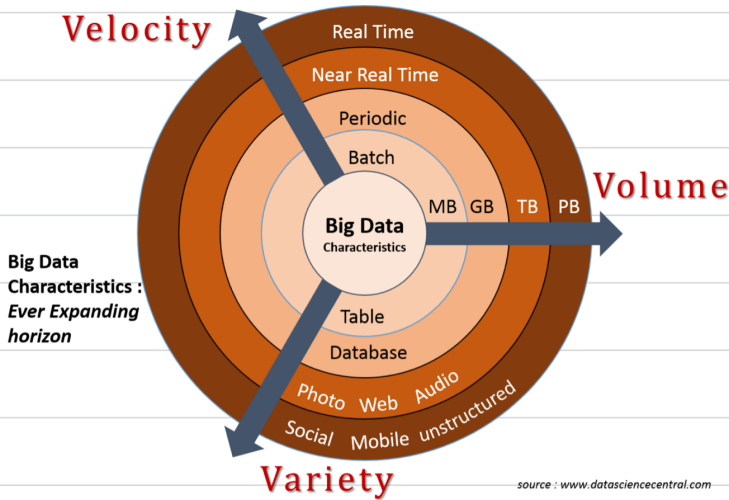
① Chunk wise processing.

100 GB CSV File
└──┬──
Load up 16 GB
Process
store, ↓

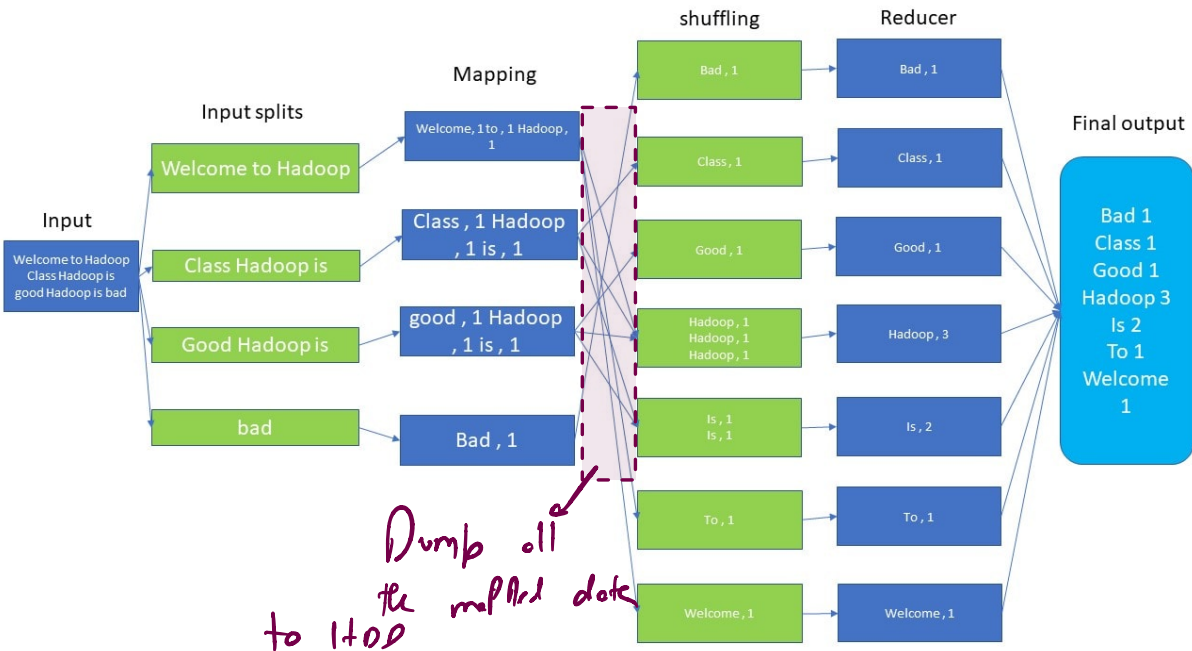
② Lazy Evaluation.

③ Distributed Computing.

Big-Data



Map Reduce Framework





Search X

For you **Trending** News Sports Enterta**India trends**

1 · Trending

#PropheciesAboutSantRampalji

129K posts

2 · Only on Twitter · Trending

#TimeManagementTips

28.2K posts

3 · Politics · Trending

सनातन धर्म

295K posts

Apache Spark

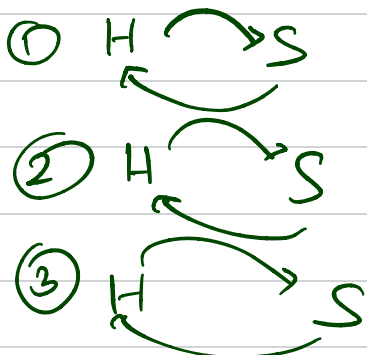
① deeps everything in RAM

② Open Source

③ Distributed Computing.

④ Lazy Evaluation.

Eager Execution



Lazy Evaluation.

Groceries

① In the End
I buy the
stuff

① Milk

② Bread

③ 2kg Potatoes

④ 1kg. Rice

$a=5$

L_1  Compiler $\rightarrow$ bytecode.

L_2

L_3

,

,

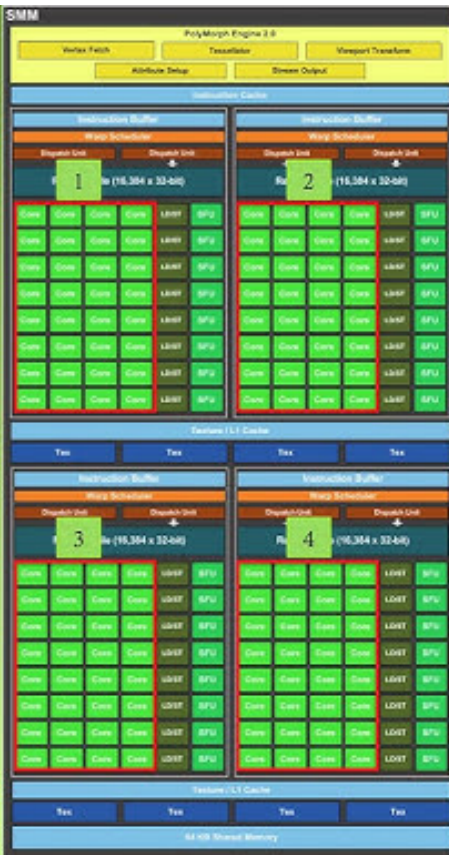
,

L_{10}

C++ $\rightarrow$ compile $\rightarrow$

g++ a.cpp

./a.out



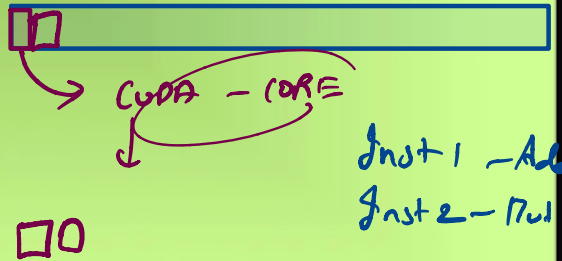
24 GB

4 x 8 = 32 cores per block

→ 32 cores/block x 4 blocks = 128 cores

128 Cores for a single SMM

256 KB — 24 GB



Number of cores in RTX 4090

Thought for a couple of seconds >

The RTX 4090 has 16,384 CUDA cores. NVIDIA's official spec lists 16,384 NVIDIA CUDA Cores.



Other core counts:

Core type	Count
CUDA cores	16,384
Tensor cores	512
RT cores	128

Usually, when people ask "number of cores" for an NVIDIA GPU, they mean **CUDA cores**.

What is a key limitation of using pandas for Big Data processing that Apache Spark addresses more effectively?

0 users have participated

- A

Pandas has more complex functions for simple data analysis tasks.

0%
- ✓ B

Pandas operates primarily on in-memory data and may struggle with datasets that exceed available RAM.

0%
- C

Pandas has a more extensive library for machine learning algorithms compared to Spark.

0%
- D

Pandas can only process data in structured formats like CSV or Excel.

0%

End Quiz Now

Based on all quizzes from the session

R

2

Rishikesh Kumar ...

1/1 ⚡ 88.03

1

Sanidhya Shukla

1/1 ⚡ 98.90

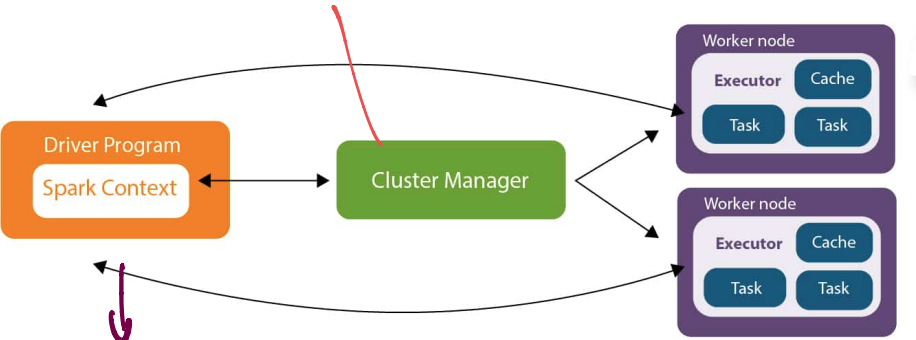
3

Manikandan A

1/1 ⚡ 86.59

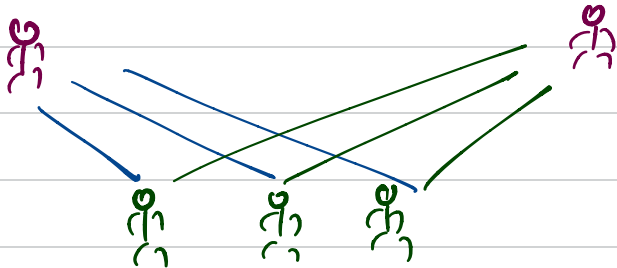
4	Kuntal Kole	1/1 ⚡ 86.19
5	K. Janani	1/1 ⚡ 83.33
6	J. Ramesh Reddy	1/1 ⚡ 82.26
7	Jigar	1/1 ⚡ 75.60

decides how much RAM/CPU should be allowed to each task.



Create and assign disk partitions

10 diff Servers



What is the primary advantage of using Apache Spark over traditional data processing tools for handling Big Data?

3 users have participated

- A

Spark can only handle structured data types

0%
- B

Spark requires a deep understanding of Java programming.

0%
- C

Spark restricts the use of machine learning libraries.

0%
- ✓

D

Spark provides fast data processing for large, complex datasets.

100%

End Quiz Now

How does PySpark handle data partitioning by default when creating an RDD?

0 users have participated

- A

PySpark keeps all data on a single partition to enhance data security.

0%
- B

PySpark does not support partitioning; all data must be managed manually.

0%
- ✓

C

By automatically determining the number of partitions based on available resources and datasets.

0%
- D

PySpark randomly scatters data across the network without regard for resource optimization.

0%

End Quiz Now

Based on all quizzes from the session

R			S			S		
Rishabh Kumar ...			Sanidhya Shukla			Kuntal Koley		
2/0			180.33			139.70		
177.85			170.70			163.70		
163.70			89.40			86.59		
86.59			83.33			79.20		
79.20			77.22					